MIL-HDBK-271-106
20 JANUARY 1986

MILITARY HANDBOOK

STRUCTURAL PANEL FASTENERS-INSTALLATION AND REMOVAL METHODS FOR MIL-F-85616, MIL-F-85617 AND MIL-F-85618



AMSC N/A

FSC 53GP

Source: https://as
Check the latest to

DISTRIBUTION STATEMENT A. Approved for public release: distribution is unlimited.

DEPARTMENT OF DEFENSE
WASHINGTON 25 DC 20361

MIL-HDBK-271-106

Structural Panel Fasteners, Installation and Removal Methods

1. This standardization handbook was developed by the Department of Defense in accordance with established procedure.

2. This publication was approved on 20 January 1986 for printing and inclusion in the military standardization handbook series.

3. This handbook provides installation and removal methods for structural panel fasteners including hole preparation and procedures for installing and removing the grommet, sleevebolt, and receptacle barrel. It will provide valuable information and guidance to mechanics at Navy, Army and Air Force Rework Facilities.

4. This handbook is not to be referenced in procurement specifications except for information purposes, nor shall it supersede any specification requirements.

5. Every effort has been made to include the latest information on installation and removal methods of panel fasteners. Mechanics should review this handbook periodically. Users of this handbook are encouraged to report any errors or recommendations to the Commanding Officer, Naval Air Engineering Center, Code 93, SESD, Lakehurst, NJ 08733.

6. The individual parts covered in this handbook should be ordered under the specification sheets associated with MIL-F-85616 (stud), MIL-F-85617 (receptacle), and MIL-G-85618 (grommet). Specifics concerning the part numbering scheme are covered in Section 4.

Source: https
Check the sourc

CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Scope	1
2.	REQUIREMENTS AND METHODS	2
2.1	Panel hole preparation	2
2.2	Substructure hole preparation	2
2.3	Grommet installation and removal.	3
2.3.1	Installing grommet into panel	3
2.3.2	Removing grommet from metal panels	4
2.3.3	Removing grommet from composite panels	5
2.4	Sleevebolt installation and removal	5
2.4.1	Installing sleevebolt into grommet	5
2.4.2	Removing sleevebolt from grommet.	6
2.5	Receptacle barrel installation and removal	7
2.5.1	Installing barrel into sub-structure	7
2.5.2	Barrel removal	8
3.	COMPLETING INSTALLATION	8
3.1	Final seating torque	8
4.	PARTS NUMBERING AND TOOLING PROCUREMENT	9
4.1	Stud number example	9
4.2	Receptacle number example	9
4.3	Grommet number example	9
4.4	Tooling procurement	9

Source: htt
Check the sou

1. SCOPE

1.1 Scope. This handbook covers the use and maintenance of structural panel fasteners. It illustrates use of the installation and removal tools, product identification, panel preparation and seating torque values. Figure 1 shows the various parts of the assembly.

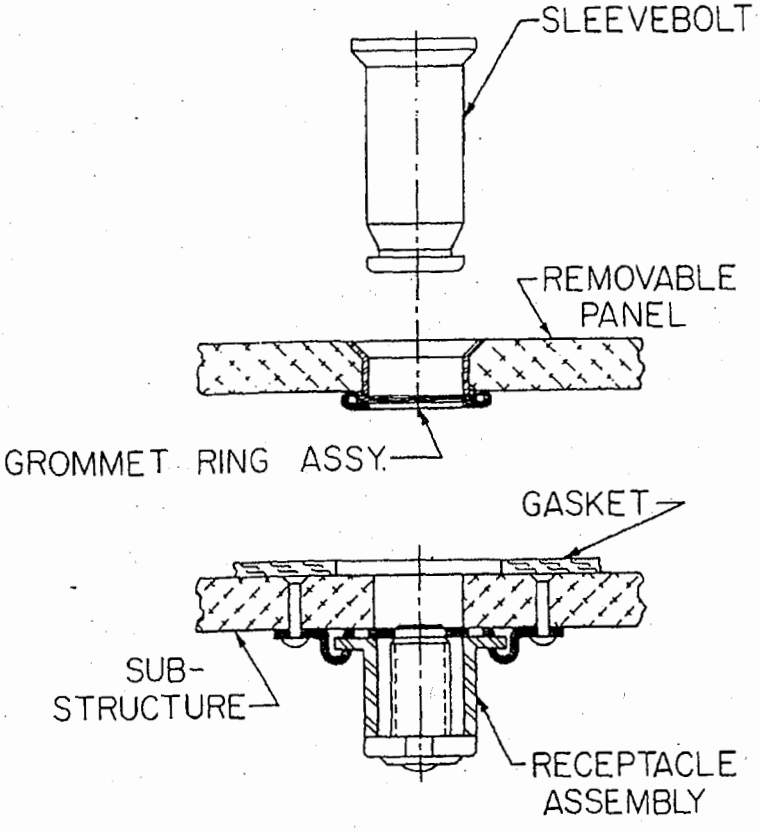


FIGURE 1. Parts of fastener assembly.

Source: https://assist
Check the source to ver

2. REQUIREMENTS AND METHODS

2.1 Panel hole preparation. Prepare the panel hole for grommet installation in accordance with Figure 2 and Table I.

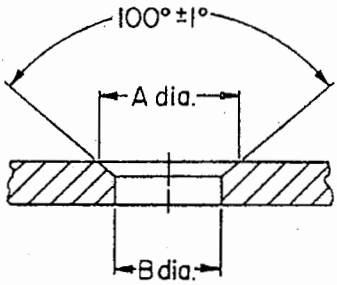


FIGURE 2. Panel hole preparation.

TABLE I. Panel hole dimensions.

Diameter Dash No.	Fastener Dia.-Nom.	A Dia.	B Dia.
-4	.250"	.403 .393	.316 .312
-5	.312"	.486 .476	.379 .375
-6	.375"	.568 .558	.472 .468

2.2 Substructure hole preparation. Prepare hole for receptacle in accordance with Figure 3 and Table II.

Source: https://assist
Check the source to ve

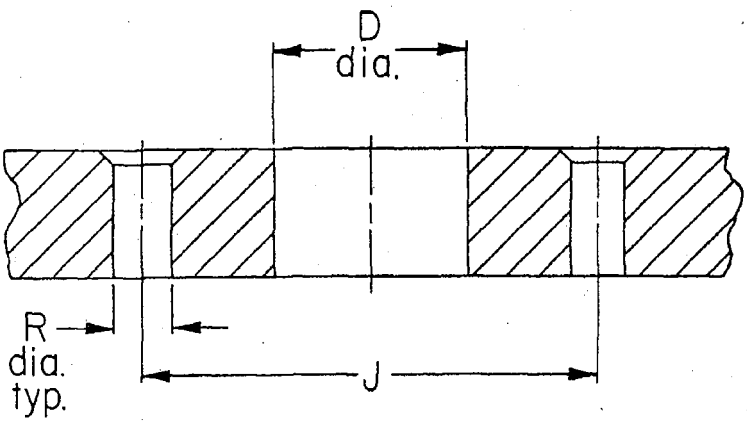


FIGURE 3. Substructure hole preparation.

TABLE 2. Substructure hole dimensions.

Diameter Dash No.	D Dia.	J	R Rivet Dia.
-4	.257 .253	.752 .748	.103 .098
-5	.318 .313	.877 .873	.103 .098
-6	.383 .377	1.002 .998	.135 .130

2.3 Grommet installation and removal.

2.3.1 Installing grommet into panel. Insert grommet shank into fastener hole from the bottom side so that large diameter of grommet is against the inside of the panel. Put threaded stud assembly of installation tool through grommet and fastener hole from the inside and engage the nut component (see Table III). "Install" end of nut should be toward the grommet. Turn nut until it contacts the grommet. Inspect to ensure that grommet and tool are properly seated, then wrench until the shank is flared to fit the countersink and the grommet is difficult to turn (70 inch pounds max). Figure 4 illustrates this procedure.

Source: https://
Check the source

TABLE III. Installation and removal tools.

Fastener Dash No.	Fastener Dia-Nom.	Tool No.*
-4	.250"	NST 130-4
-5	.312"	NST 130-5
-6	.375"	NST 130-6

*See 4.4.

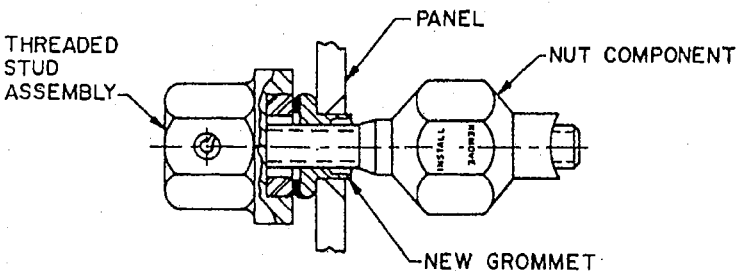


FIGURE 4. Grommet installation.

2.3.2 Removing grommet from metal panels. Insert proper threaded stud assembly through grommet from the inside and engage the nut component (see Table III) with "Remove" end toward grommet. Hand turn nut to engage with grommet, then wrench the hexagon components of both the nut and the stud assembly until the grommet shank is severed. A sudden easing of wrench torque will be felt when the grommet breaks. Disassemble tool and remove debris.

Source: https
Check the sour

2.3.3 Removing grommet from composite panels. Insert pilot of removal tool through grommet from countersunk side (see Figure 5 and Table IV). Push down knurled sleeve and restrain from rotating. Turn handle clockwise until grommet shank severs.

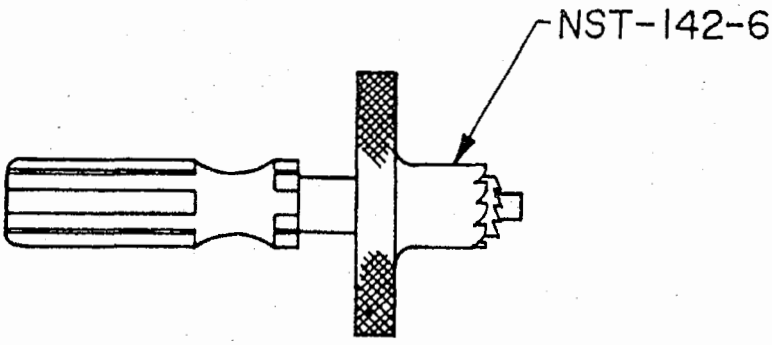


FIGURE 5. Grommet removal tool for composite applications.

TABLE IV. Grommet removal tools for composite applications.

First Dash No. of Grommet	Fastener Dia-Nom.	Tool No.*
-4	.250"	142-4
-5	.312"	142-5
-6	.375"	142-6

*See 4.4.

2.4 Sleevebolt installation and removal.

2.4.1 Installing sleevebolt into grommet. Sleevebolt is fitted into the installed grommet ring assembly using finger pressure and a slight rocking motion to engage ring.

Source: https://assist.dla.mi
Check the source to verify tha

2.4.2 Removing sleevebolt from grommet. Engage threaded stud of NST-132 tool (see Table V) into sleevebolt to its full depth and hand tighten. Engage hexagon driver to prevent sleevebolt rotation. Turn the NST-132 tool by wrenching on its hexagon two or three rotations until the tool cuts the lip of the sleevebolt away. After the tool is removed, the sleevebolt may be pushed from the grommet. This alignment is illustrated by Figure 6. A new sleevebolt may now be installed from the top side of the panel.

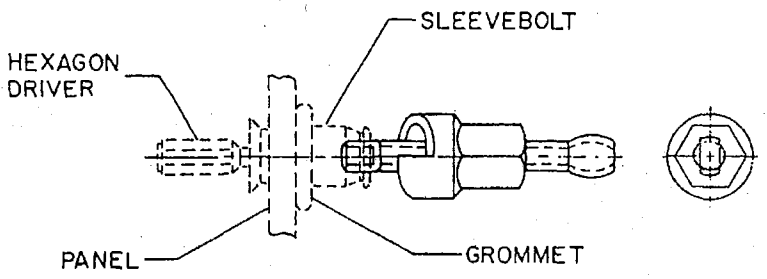


FIGURE 6. Sleevebolt removal.

TABLE V. Sleevebolt removal tools.

Fastener Dash No.	Fastener Dia-Nom.	Tool No.*
-4	.250"	NST132-4
-5	.312"	NST132-5
-6	.375"	NST132-6

*See 4.4.

Source: http
Check the sou

2.5 Receptacle barrel installation and removal.

2.5.1 Installing barrel into substructure. Insert one lug into shell which has been riveted to the substructure. Strike the opposite lug with a screwdriver as shown in Figure 7. An alternate method is to insert barrel assembly into installation tool (see Table VI), engage one lug in shell and rock tool until barrel assembly snaps into place.

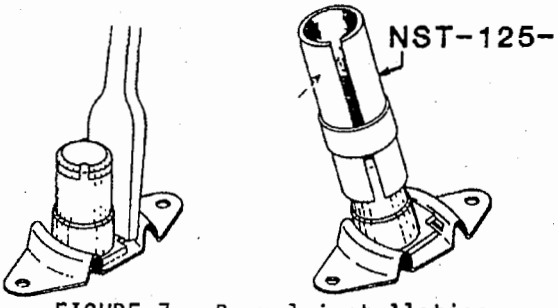


FIGURE 7. Barrel installation.

TABLE VI. Barrel installation and removal tools.

Receptacle Assy No.		Removal Tool Part Number*	Installation Tool Part Number*
First Dash Number	Fastener Dia.-Nom.		
-4	.250"	NR-2	NST-125-4
-5	.312"	NR-1	NST-125-5
-6	.375"	NR-3	NST-125-6

*See 4.4.

Source: https://
Check the source

2.5.2 Barrel removal. Insert removal tool (as specified in Table VI) with downward pressure, between receptacle barrel assembly and shell. See Figure 8. Press against barrel assembly to insure engagement. Barrel assembly will snap out with a simple prying motion.

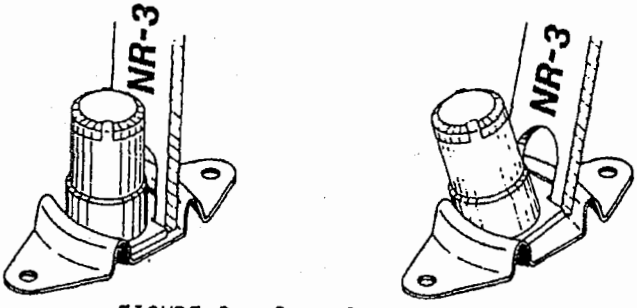


FIGURE 8. Barrel removal.

3. COMPLETING INSTALLATION

3.1 Final seating torque. To complete installation of the fastener, apply the torque specified in Table VII.

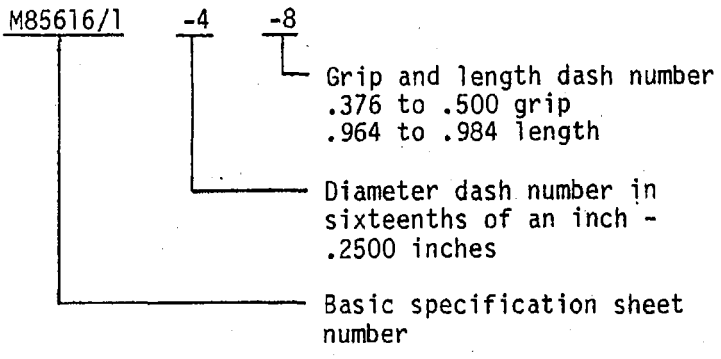
TABLE VII. Final seating torque.

First Dash Number	Fastener Dia.-Nom.	Torque Inch-Pounds
-4	.250"	30-40
-5	.312"	40-50
-6	.375"	95-110

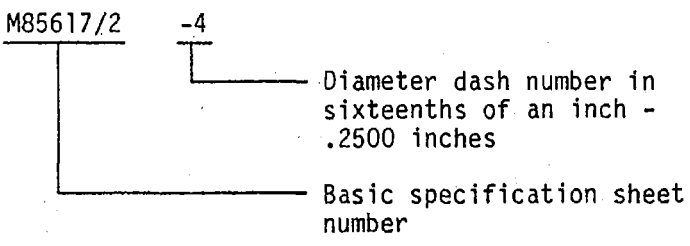
Sourc
Check th

4. PARTS NUMBERING AND TOOLING PROCUREMENT

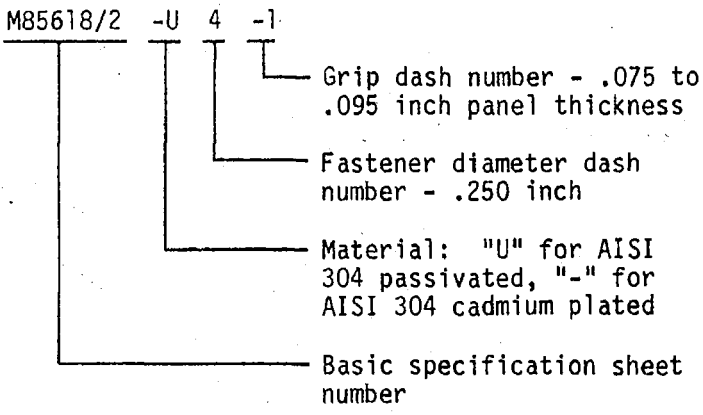
4.1 Stud number example.



4.2 Receptacle number example.



4.3 Grommet number example.



4.4 Tooling procurement. The tools described and referenced in this handbook are available from:

SPS Technologies
2701 South Harbor Boulevard
Santa Ana, CA 92707

Source: https://assist.
Check the source to ver

Custodians:
Navy - AS
Air Force - 11
Army - AV

Preparing Activity:
Navy - AS
Project No. 53GP-0149-07

Source: http
Check the sou